SOFTWARE DEVELOPMENT PROJECT SPECIFICATION

Course: ISP392 - Software Development Project

Project Title: HEAVENSTAY WELLNESS RESORT MANAGEMENT SYSTEM

Project Acronym: HSWRMS

Team Structure: 6 Students / 6 Modules

Domain Application: Hospitality, Wellness Tourism, Healthcare, E-Commerce

Revision: HSWRMS Project Specification Ver2 - October 2026

> This revision updates the September 2026 baseline to align the Project Specification with the current SRS, Business Flows, Use Case Diagrams, Screen Flow and agreed module behavior.

1. Problem Statement

Traditional Property Management Systems mainly support standard hotel operations such as room booking, check-in and checkout. A wellness resort requires a broader journey combining retreat packages, accommodation, sensitive health and dietary information, Spa/Therapy/Yoga scheduling, personalized dining, extra services, consolidated billing and management reporting.

HEAVENSTAY WELLNESS RESORT MANAGEMENT SYSTEM (HSWRMS) is an integrated system for a fictional wellness resort. It supports the Customer journey from registration and retreat booking through check-in, treatment scheduling, personalized F&B;, checkout, payment, review and reporting.

HSWRMS must:

- prevent scheduling conflicts for both Therapist/Yoga Instructor and Treatment Room;
- protect health, allergy, dietary and residence-identification data using role-based access;
- automatically coordinate Spa/Therapy scheduling and recovery while keeping final Customer/Receptionist confirmation;
- use ingredient batch information to determine F&B; and menu availability;
- consolidate Package, Spa, F&B; and approved Villa/Facility charges into one Customer Folio;
- prevent checkout while required charges or operational checks remain incomplete;
- separate payment confirmation from checkout completion;
- provide management reporting without granting Manager unrestricted Admin authority.

2. Third-Party Tool Interfaces

HSWRMS integrates with the following external services.

2.1 VNPay Payment Gateway

Purpose: Process supported online payment transactions for retreat deposits and final checkout.

Main interactions:

- HSWRMS sends payment amount and transaction/order information to VNPay.
- VNPay returns the payment result to HSWRMS.

- HSWRMS records the confirmed transaction against the related booking or invoice.
- A successful payment transaction does not by itself complete resort checkout.

2.2 Calendar & Notification Services

Purpose: Support account notifications and Spa/Therapy session communication.

Main interactions:

- send email OTP for account verification;
- synchronize confirmed or changed Spa/Therapy sessions to the Customer calendar;
- send T-24h informational reminder;
- send T-2h primary confirmation request;
- send T-75m final reminder when no response has been received;
- support the T-60m confirmation deadline;
- send T-15m upcoming-session reminder for an active confirmed session;
- notify Customers of accepted schedule or Therapist assignment changes.

HSWRMS remains the source of truth for scheduling decisions and session status.

2.3 Identity Providers

Google Identity and Facebook Login

Purpose: Provide Social Single Sign-On (SSO) in addition to normal email/password authentication.

3. External Entity Specification (Actors)

1. Guest: An unauthenticated public user who can browse public resort/retreat information and start registration, sign-in, password recovery or supported SSO. Guest cannot access authenticated Customer or staff functions.
2. Registered User: Abstract/general actor representing an authenticated HSWRMS account holder. It provides common authentication/account interactions inherited by Customer and staff roles; it is not an independent business role and grants no additional permission by itself.
3. Customer: Authenticated resort customer who manages personal wellness information, books retreat services, confirms Spa/Therapy schedules, selects meals, views itinerary/folio, makes supported payments and may review a completed stay.
4. Receptionist: Handles arrivals, check-in/check-out, residence information, physical Villa assignment/status, Customer folio operations and Spa/Therapy requests on behalf of Customers. Receptionist does not manually choose Therapist or Treatment Room.
5. Spa Therapist / Yoga Instructor: Views assigned schedule and the Customer's Treatment Safety Health Profile required for safe treatment, performs the assigned session and records the final attendance outcome. Unrelated payment, residence and account-security information remains hidden.
6. F&B; Staff / Chef: Manages meal-preparation operations, ingredient batch/availability information, necessary allergy/dietary warnings, ingredient usage and meal-order status without accessing unrelated medical information.
7. Manager: Uses approved read-only/restricted dashboards and reports such as revenue, occupancy and Therapist utilization. Manager does not have unrestricted Admin authority by default.
8. Admin: Manages staff accounts, approved master data, system parameters, feature flags, permission configuration, limited delegation and audit/security administration.

External Systems

- VNPay Payment Gateway
- Calendar & Notification Services
- Google Identity
- Facebook Login

4. Use Cases / User Stories

The project is organized into 6 interconnected modules. Each module has a primary owner while sharing data and business rules across the system.

Module 1 - Authentication & Sensitive Health Profile

- UC01 - Register, Verify Email & Login: As a Guest, I want to create an account, verify email, sign in securely or use supported Google/Facebook SSO.
- UC02 - Manage Dietary & Health Profile: As a Customer, I want to create and maintain dietary, allergy and treatment-safety information with explicit consent.
- UC03 - Manage Staff Accounts & Roles: As an Admin, I want to create, edit, deactivate staff accounts and assign authorized roles.
- UC04 - Manage Master Data: As an Admin, I want to manage approved system master data such as Villa Categories, Retreat Packages, Spa Services and other configured catalogs used by operations.
- UC05 - Request Health Data Deletion: As a Customer, I want to request deletion/anonymization of eligible health/allergy data after the retreat, subject to required retention and audit rules.

Module 2 - Retreat Package & Accommodation Booking

- UC06 - Browse Retreat Packages: As a Guest or Customer, I want to browse available Retreat Packages and public package details.
- UC07 - Book Retreat & Pay Deposit: As a Customer, I want to select a Retreat Package, dates and Villa type, then pay the required deposit through VNPay.
- UC08 - Check-In & Assign Villa: As a Receptionist, I want to verify the Customer, collect required residence-identification information and assign an available physical Villa.
- UC09 - Manage Villa Status: As a Receptionist, I want to manage Villa operational states such as Available, Occupied, Maintenance and Needs Cleaning.
- UC10 - View Booking & Itinerary: As a Customer, I want to view my booking, accommodation information and itinerary.

Residence-compliance clarification: Required Customer identification/residence data must be protected at rest and limited to authorized access.

Module 3 - Spa & Therapy Scheduling Engine - CORE

- UC11 - Schedule Spa/Therapy Session: As a Customer or Receptionist, I want to request a Spa/Therapy service and confirm one of the scheduling options automatically generated by HSWRMS.
- UC12 - Automatic Therapist & Treatment Room Matching: HSWRMS generates ranked candidates using preferred day/time constraints and confirms a candidate only when a qualified Therapist/Yoga Instructor and compatible Treatment Room are both available for the full interval.

- UC13 - Manage Pre-Session Confirmation & Recovery: HSWRMS manages reminders, Customer confirmation, automatic rescheduling/re-matching, schedule-change notification and slot-swap fallback when required.
- UC14 - View Therapist Schedule & Treatment Safety Health Profile: As a Therapist/Yoga Instructor, I want to view assigned sessions and the treatment-safety health information required to assess contraindications and safely deliver treatment.
- UC15 - Record Session Attendance & Outcome: As the assigned Therapist/Yoga Instructor, I want to record the final session outcome as Completed or No-Show.

Supporting behavior:

- Customer/Receptionist may provide treatment and preferred day/time constraints but does not manually choose Therapist or Treatment Room.
- HSWRMS revalidates the selected candidate before commit and reserves Therapist + Treatment Room atomically.
- Extra Spa/Therapy services use the same scheduling engine and applicable charges are posted to the Customer Folio.
- Confirmation timeline: T-24h information, T-2h primary confirmation, T-75m final reminder, T-60m deadline and T-15m upcoming reminder.
- Declined/Expired confirmation and Therapist unavailability trigger automatic recovery.
- Customer slot swap is a final fallback and requires explicit acceptance from every affected Customer before revalidation and atomic commit.
- Customer confirmation is not attendance; the Therapist/Yoga Instructor owns the final Completed/No-Show outcome.

Module 4 - Dietary F&B; Management - CORE

- UC16 - Manage Ingredient Batches & Availability: As F&B; Staff/Chef, I want to maintain ingredient batch information including food type, received date, expiry date, quantity and condition so HSWRMS can determine ingredient availability.
- UC17 - Select Personalized Daily Meals: As a Customer, I want to view and select available meals filtered by relevant allergy/dietary information and current menu availability.
- UC18 - Order Extra A-la-carte Items: As a Customer, I want to order additional F&B; items and post the applicable charge to my Customer Folio.
- UC19 - Fulfill Meal Orders: As F&B; Staff/Chef, I want to view required allergy/dietary alerts, prepare meals, record ingredient usage and update order status through Preparing → Ready → Delivered.
- UC20 - Enforce F&B; Health-Data Minimization: HSWRMS exposes only food-related allergy/dietary information required for meal preparation and hides unrelated medical history.

Supporting behavior:

- HSWRMS recalculates ingredient and related menu availability based on quantity, expiry and condition.
- Ingredient usage updates remaining quantity and may change menu availability.
- Daily meals and extra a-la-carte items use the same meal-order fulfillment process where applicable.

Module 5 - Consolidated Checkout & Analytics

- UC21 - Inspect Villa & Finalize Consolidated Invoice: As a Receptionist, I want to inspect the Villa/facility checklist, record chargeable missing/damaged items when applicable, and finalize one Customer invoice containing Package, Spa, F&B; and approved Villa/Facility charges.

- UC22 - Process Final Payment: As a Receptionist, I want to process the outstanding final amount using supported payment methods such as Cash or VNPay and record the confirmed payment result.
- UC23 - Complete Checkout & Submit Review: As a Receptionist, I want to complete checkout only after all requirements are satisfied; after checkout, the Customer may optionally submit a review and rating.
- UC24 - View Revenue Dashboard: As a Manager, I want to view revenue breakdown across Retreat Package, Spa/Therapy, F&B; and other approved charge categories.
- UC25 - View Occupancy & Therapist Utilization Reports: As a Manager, I want to view permitted occupancy and Therapist-utilization information and export supported report data where available.

Checkout rules:

- Villa/facility inspection occurs before the final invoice is locked.
- Approved Villa/Facility charges are posted to the Customer Folio before final payment.
- Checkout is blocked while required Spa, F&B;, Villa/Facility or other outstanding charges remain unresolved.
- Payment confirmation does not equal checkout completion.
- After checkout completion, the Villa status is updated to the configured post-checkout state such as Needs Cleaning.
- Review/rating is optional and available only after a completed stay.

Module 6 - System Administration & Permission Management

- UC26 - Configure Business Parameters: As an Admin, I want to configure approved system parameters such as cancellation windows, refund percentages and other supported rules.
- UC27 - Toggle Feature Flags: As an Admin, I want to enable or disable approved functionality by module.
- UC28 - Configure Permission Matrix: As an Admin, I want to configure detailed permissions by role, screen and action.
- UC29 - Delegate Limited Admin Rights: As an Admin, I want to grant or revoke a limited subset of administrative capabilities for Manager without granting full Admin authority.
- UC30 - View Sensitive-Data Audit Log: As an Admin, I want to review sensitive-data access and security-relevant events.
- UC31 - View System Overview Dashboard: As an Admin or authorized Manager, I want to view system-level operational information according to granted access.

5. Business Rules (Strict Constraints)

1. BR-01 - Guest vs Customer State: Guest is unauthenticated and limited to public/authentication functions. Authenticated resort business functions are performed by Customer or authorized staff roles.
2. BR-02 - Automated Spa Scheduling: HSWRMS generates scheduling candidates; Customer/Receptionist does not manually select Therapist or Treatment Room.
3. BR-03 - Joint Availability, Revalidation & Atomic Commit: A Spa/Therapy candidate is valid only when a qualified Therapist/Yoga Instructor and compatible Treatment Room are available for the same complete interval. Availability must be revalidated immediately before atomic reservation/commit.
4. BR-04 - Spa Confirmation, Recovery & Swap Consent: Declined/Expired confirmation or Therapist unavailability triggers automatic recovery. Any confirmed Customer time change requires Customer acceptance. A slot swap is committed only after every affected Customer explicitly accepts and the proposal is revalidated.
5. BR-05 - Confirmation Is Not Attendance: Pre-session Customer confirmation only retains the slot. Only the assigned Therapist/Yoga Instructor records final attendance as Completed or No-Show.

6 BR-06 - Treatment Safety Data Access: Therapist/Yoga Instructor may access the complete treatment-safety health information required for assigned treatment, while unrelated administrative, residence, payment and account-security data remains hidden.

7 BR-07 - F&B; Data Minimization & Availability: F&B; Staff/Chef may access only required food-related allergy/dietary information. Ingredient batch quantity, expiry and condition determine ingredient and related menu availability.

8 BR-08 - Consolidated Folio Accounting: Package, Extra Spa/Therapy, Extra F&B; and approved Villa/Facility charges are posted to the Customer Folio with traceable links to the related stay/booking.

9 BR-09 - Checkout Guard: Checkout cannot complete while required charges, payment confirmation or required checkout checks remain unresolved.

10 BR-10 - Payment vs Checkout: A successful payment transaction records payment only; HSWRMS completes checkout separately after all checkout conditions are satisfied.

11 BR-11 - Residence Data Protection: Residence-identification data collected at check-in must be protected at rest and accessible only to authorized roles.

12 BR-12 - Manager Least Privilege: Manager may access approved dashboards/reports and explicitly delegated limited capabilities but must not inherit unrestricted Admin permissions.

13 BR-13 - Health Data Deletion Timing: Customer health/allergy deletion or anonymization is available subject to retreat lifecycle, operational retention and audit requirements.

6. References & Domain Knowledge (Mandatory Research)

Students must continue to research and document the legal, hospitality, wellness and technical basis used by the project.

A. Legal Compliance & Data Protection

1 Vietnam personal-data protection requirements (as referenced by the supplied project baseline)

Focus areas: sensitive health data, explicit consent, access control and appropriate protection. Project application: Health/Diet data requires explicit consent, must not use default-checked consent and must be protected appropriately.

1 Law on Residence 2020 (Luật Cư trú 2020)

Focus area: temporary-residence information for resort Customers. Project application: check-in collects required residence-identification information; access is restricted and protected at rest.

B. Hospitality & Wellness Domain Operations

1 Global Wellness Institute (GWI) Standards / Wellness Taxonomy

Project application: Retreat Package and wellness-service terminology should reflect the wellness-tourism domain rather than generic hotel wording.

1 Guest/Customer Folio & Night Audit Practices (AHLEI-oriented research)

Project application: operational service charges are posted to a central Customer Folio and consolidated during checkout.

C. Technical Synthesis

1 Encryption, Authorization and Concurrency Research

Focus areas: protection of sensitive data at rest, backend RBAC, transactional resource reservation, scheduling concurrency, payment confirmation and auditability.

7. Module Integration Summary

The six modules form one end-to-end resort journey:

> Authentication & Wellness Profile → Retreat Booking & Deposit → Check-In & Villa Assignment → Spa/Therapy + F&B; Operations → Customer Folio & Villa Inspection → Final Payment → Checkout → Review & Analytics

Module 6 operates across the system to control permissions, configuration, auditability and management visibility.

Key integrations:

- M1 <→ M6: account roles, permission model and controlled administration;
- M2 / M5 <→ VNPay: retreat deposit and supported final online payment;
- M3 <→ Calendar & Notification Services: calendar synchronization, confirmation lifecycle and schedule-change notifications;
- M2 + M3 + M4 + Villa/Facility inspection → M5: charges are aggregated into the Customer Folio and final invoice;
- M4: ingredient usage updates inventory availability and may affect Customer-facing menu availability.

End of HSWRMS Project Specification Ver2